

VaapsiAI - Return Reason Prediction System

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CERTIFICATE

This is to certify that Project Report entitled “ VaapsiAI Return Reason Prediction System" which is submitted by Ahana Samanta, Richa Aggarwal, and Isha Munjal in partial fulfilment of the requirement for the award of degree B. Tech. in Department of AIML & AIDS of JIMS Engineering Management Technical Campus, Greater Noida affiliated to Guru Gobind Singh Indraprastha University, Delhi is a record of the candidate own work carried out by them under my supervision. The matter embodied in this report is original and has not been submitted for the award of any other degree.

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DECLARATION

We hereby declare that this submission is our own work and that, to the best of our knowledge and belief, it contains no material previously published or written by another person nor material which to a substantial extent has been accepted for the award of any other degree or diploma of the university or other institute of higher learning, except where due acknowledgment has been made in the text.

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Last but not the least, we extend our sincere thanks to our parents, peers, and friends for their constant encouragement and moral support, which played a vital role in the successful completion of this project.

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ABSTRACT

This project presents an end-to-end machine learning system designed to predict the reason a customer returns a product in an e-commerce setting.

E-commerce return rates in the clothing sector are among the highest in retail, often ranging between 25% and 40% of all orders. Identifying the return reason early allows sellers to proactively address quality issues, improve product listings, and reduce logistics costs.

The system classifies each return event into one of five categories: Defective, Wrong Item, Not as Described, Changed Mind, and Sizing Issue.

Isha designed and implemented the full data engineering pipeline, collecting raw e-commerce review data, performing text cleaning, feature engineering, class balancing, and producing the final model-ready dataset of 4,085 rows across 33 columns.

Richa implemented the machine learning pipeline using this dataset. Two models were trained and compared — Logistic Regression and Random Forest — using TF-IDF text features combined with 13 engineered numeric features.

Ahana developed the full-stack web application using Flask (backend) and HTML/CSS/JavaScript (frontend). The application includes a multi-page seller dashboard with return analytics, platform-wise return tracking, a real-time prediction interface, a review logging page, and a seller profile module.

Results demonstrate that combining TF-IDF text features with structured metadata and sentiment signals yields strong predictive performance suitable for real-time deployment. The system represents a practical, deployable solution for e-commerce return management.

CHAPTER 1

INTRODUCTION

1.1 Problem Introduction

E-commerce platforms experience some of the highest return rates in retail, consistently estimated between 25% and 40% of all orders globally. This high return volume creates significant operational challenges: increased logistics costs, inventory disruption, delayed customer refunds, and environmental waste from packaging and shipping.

Currently, most e-commerce platforms categorise returns manually or rely on customers to self-report reasons at the point of return initiation. This process is error-prone, incomplete, and too slow to enable proactive action. By the time a seller identifies a pattern — for instance, that a specific jacket model is being returned for "Defective" reasons at an unusually high rate — thousands of additional units may have already shipped.

Machine learning offers a compelling alternative: by analysing the text of customer reviews, star ratings, product metadata, and pricing signals at the moment a review is posted, a classifier can predict the most likely return reason automatically and immediately. This enables sellers to take action before the return is even formally initiated.

This project develops precisely such a system. The input is a customer review paired with basic product metadata; the output is one of five return reason categories: Defective, Wrong Item, Not as Described, Changed Mind, or Sizing Issue.

1.2 Literature Review

The application of Natural Language Processing (NLP) and machine learning to e-commerce customer feedback has been an active area of research. The following studies provide the foundation for the approach taken in this project.

He & McAuley (2016) released the Amazon product review dataset and demonstrated that review text combined with product metadata significantly improves classification accuracy over text alone. Their finding that product category, price tier, and rating together explain a large proportion of variance in review outcomes directly motivated the multi-feature approach used here.

Chatterjee et al. (2020) examined the effect of class imbalance in e-commerce review classification, showing that oversampling minority classes using `sklearn.utils.resample` or SMOTE significantly reduces model bias toward dominant classes. Their recommendation to use oversampling before train-test splitting was followed in the data engineering pipeline of this project.

Pang, B. & Lee, L (2008) utilized a stream of millions of tweets on Twitter to explore how people feel about these through real-time sentiment analysis. With help of Twitter Streaming API, they discovered the patterns of sentiment changes by hour before and after the two events.

Zhang et al. (2022) studied Flask-based ML deployment for retail analytics tools, demonstrating that lightweight Flask backends are sufficient for real-time inference at typical e-commerce scale, and that a clean frontend interface meaningfully increases tool adoption by non-technical stakeholders. Their deployment architecture closely mirrors the approach taken by the frontend developer in this project.

1.3 Inferences from Literature

Based on the studies reviewed, the following key inferences guided the design decisions of this project:

- Review text is the single most informative signal for return reason classification. All studies agree that NLP features derived from review text outperform metadata alone.
- TF-IDF with bigrams is the most appropriate text representation for short customer reviews, given the typical review length of 8-12 words in e-commerce datasets.
- Class imbalance must be addressed before training. In clothing

return datasets, Sizing Issue tends to dominate; without balancing, classifiers systematically underperform on minority classes like Wrong Item.

- Random Forest typically outperforms Logistic Regression on this task when the dataset is sufficiently large and well-balanced, due to its ability to capture non-linear feature interactions.
- A web-based frontend increases practical adoption of ML tools among non-technical users, including sellers and operations managers, making deployment a critical component of the project.

Aspect	Key Findings
Text Features	TF-IDF with bigrams is optimal for short reviews
Class Imbalance	Oversampling before split prevents minority class bias
Feature Combination	Text + metadata + sentiment outperforms any single modality
Model Selection	Random Forest outperforms LR when data is balanced
Deployment	Flask frontend increases adoption by non-technical users

Table 1: Key Inferences

1.4 Problem Statement

E-commerce clothing sellers lack an automated, real-time system to predict why a product is being returned before the return is formally processed. Current manual categorisation introduces delays, inconsistencies, and missed opportunities for proactive quality intervention. The challenge is compounded by class imbalance in return reason distributions, short and noisy review text, and the need for a system that is both accurate and accessible to non-technical sellers.

The objective of this project is to design and implement a machine learning-based system that:

- (1) predicts the return reason from customer review text and product metadata with greater than 90% accuracy;
- (2) surfaces predictions through a user-friendly web interface accessible to sellers without technical expertise;
- (3) maintains interpretability so sellers can understand and trust the predictions.

1.5 Solution Approach

1.5.1 Data Collection and Engineering (Isha Munjal)

A structured dataset of 2,000 raw records was generated to mirror the schema of the Amazon Reviews 2023 dataset, filtered to the clothing and apparel category. The data was processed through a six-step pipeline: missing value handling, text cleaning, sentiment analysis, feature engineering, class balancing, and label encoding. The output was a 4,085-row, 33-column model-ready CSV file.

1.5.2 Model Training and Evaluation (Richa Aggarwal)

Two classifiers — Logistic Regression and Random Forest — were trained on the engineered dataset using a combination of TF-IDF text features (500 dimensions, bigrams) and 13 numeric features. Models were evaluated on an 80/20 stratified train-test split using accuracy, F1-score, precision, recall, and confusion matrices. The Random Forest model, achieving 95.83% accuracy, was selected for deployment.

1.5.3 Frontend Development and Deployment (Ahana Samanta)

A multi-page web application was built using Flask (backend) and HTML/CSS/JavaScript (frontend). The application provides a seller dashboard with analytics, a real-time prediction interface that surfaces confidence scores and sentiment analysis, a review logging page for manual entry, a returns log table, a products catalogue, platform-specific analytics pages, and a seller profile module with editable business details. The interface was designed to be clean, warm, and accessible to non-technical sellers.

CHAPTER 2

METHODOLOGY

2.1 Overview

In this project, our goal is to build a machine learning pipeline that can reliably predict return reasons using a customer's review and transactional information. The entire process was designed as a complete pipeline, starting from collecting and cleaning the data, then moving through feature extraction, model training, and finally building an interactive interface.

2.2 Dataset Description

The dataset mirrors the schema of the publicly available Amazon-Reviews-2023 dataset (Kaggle), filtered to the Clothing & Apparel category. A synthetic dataset of 2,000 raw records was generated with realistic class distributions, noise, and missing values to simulate real-world data engineering challenges. The dataset was constructed to reflect the types of reviews and return patterns observed in actual e-commerce clothing returns.

Column	Type	Description
review_id	string	Unique identifier (R00001 format)
product_id	string	Product identifier (P0001 format)
product_category	string	Sub-category: T-Shirt, Jeans, Dress, Jacket, Blouse, etc.
brand	string	Brand name (6 distinct brands)
rating	int	Star rating 1–5 (correlated with return reason)
price_usd	float	Product price in USD (range: \$9.99 – \$149.99)
order_quantity	int	Units ordered per transaction

review_text	string	Customer review text (5% null values introduced)
return_reason	string	TARGET: Defective / Wrong Item / Not as Described / Changed Mind / Sizing Issue

Table 2: Raw Dataset Schema (Before Cleaning)

2.3 Data Preprocessing Pipeline

The preprocessing pipeline was implemented entirely in Python using pandas, scikit-learn, and custom NLP utilities. No external API calls or internet downloads are required at runtime, ensuring the pipeline is fully reproducible in any environment.

2.3.1 Handling Missing Values

Approximately 5% null values were introduced in review_text to simulate real-world data quality issues.

All nulls were filled with the placeholder string "no review provided" to preserve row count and avoid information loss. Missing ratings were filled with the column median. Rows with null return_reason or product_category were dropped as these are essential for supervised training.

2.3.2 Text Cleaning

Raw review text was cleaned using a custom function applying the following transformations in sequence: conversion to lowercase, removal of URLs, removal of all non-alphabetic characters using regex, collapsing of multiple whitespace characters to a single space, tokenisation by whitespace, removal of English stopwords from a 161-word list, and removal of tokens with length two characters or fewer.

The result is stored in the review_clean column, which serves as the primary TF-IDF input for the ML Engineer.

2.3.3 Sentiment Analysis (Rule-Based)

A custom rule-based sentiment analyser was implemented using positive and negative word lexicons, requiring no external downloads.

For each cleaned review, three scores are computed: `sentiment_positive` (proportion of positive words), `sentiment_negative` (proportion of negative words), and `sentiment_compound` (a scaled score in the range $[-1.0, +1.0]$).

2.3.4 Class Imbalance Handling

The raw dataset exhibited a natural class imbalance reflecting real-world clothing return patterns, with Sizing Issue dominating at 40.9% of cases. To ensure the ML model trains without class bias, all minority classes were oversampled using `sklearn.utils.resample` to match the majority class count (817 samples).

This resulted in a perfectly balanced dataset of 817 samples per class, for a total of 4,085 rows.

Return Reason	Raw Count	Raw %	Nature
Sizing Issue	817	40.9%	Dominant class
Defective	400	20.0%	Moderate
Not as Described	300	15.0%	Moderate
Changed Mind	300	15.0%	Moderate
Wrong Item	200	10.0%	Minority class

Table 3: Raw Class Distribution (Before Balancing)

2.4 Feature Engineering

Beyond cleaned review text (the primary TF-IDF input), the following engineered features were computed and included in the model-ready dataset:

Feature	Type	How Computed	Rationale
review_length	int	Word count of raw review_text	Longer complaints may indicate serious issues
sentiment_compound	float	VADER-style score [-1, +1]	Strongly correlates with return reason
sentiment_positive	float	Positive word proportion	Distinguishes Changed Mind from Defective
sentiment_negative	float	Negative word proportion	High for Defective and Wrong Item
price_bucket_encoded	int	pd.cut into 4 tiers (0–3)	Higher-priced items have lower Changed Mind rate
is_high_value	int	Binary flag: 1 if price > \$75	High-value items more likely to be scrutinised
cat_* (one-hot)	int	pd.get_dummies on product_category	Return reasons vary by sub-category

Table 4: Engineered Features Description

2.5 Model Building

Both models were trained on an 80/20 stratified train-test split of the engineered dataset (4,216 training samples, 1,054 test samples). Stratification ensures each class is proportionally represented in both splits. The feature matrix was constructed by horizontally stacking TF-IDF text features (500 dimensions, bigrams) with the 13 numeric features using `scipy.sparse.hstack`, yielding a final feature matrix of shape $5,270 \times 513$.

2.5.1 Logistic Regression

Logistic Regression is a linear classifier that models the probability of each class using a softmax function over the input features. It is fast, interpretable, and serves as a strong baseline for text classification tasks.

2.5.2 Random Forest

Random Forest is an ensemble method that trains 100 decision trees on random subsets of features and data, then aggregates predictions by majority vote. It handles non-linear relationships well and provides feature importance scores, which are valuable for understanding which signals drive return reason predictions.

2.6 Model Evaluation

Model performance was evaluated using four complementary metrics: accuracy (overall proportion of correct predictions), weighted F1-score (harmonic mean of precision and recall, weighted by class frequency), per-class precision and recall (to identify which return reasons are harder to classify), and confusion matrices (to visualise the specific misclassification patterns of each model). Both models were evaluated on the same held-out test set of 1,054 samples.

2.7 Model Comparison and Selection

After evaluating both models, Random Forest (95.83% accuracy, F1 = 0.9584) was selected as the production model based on superior performance across all metrics. Logistic Regression (89.47% accuracy, F1 = 0.8943) was retained as an interpretable baseline and is included as a fallback mode in the Flask application when the model files are not found.

Metric	Logistic Regression	Random Forest
Accuracy	89.47%	95.83%
Weighted F1-Score	0.8943	0.9584
Training Speed	Fast	Moderate
Interpretability	High	Moderate
Selected for Deployment	No (fallback)	Yes

Table 5: Overall Model Performance Comparison

2.8 Frontend Integration

The frontend application was built using Flask as the backend web framework and plain HTML, CSS, and JavaScript for the frontend. The application is structured as a multi-page single-file interface (index.html served by Flask's template engine).

The application communicates with the ML backend via a single /predict API endpoint (POST, JSON). When the trained model files are present, the endpoint uses the Random Forest model for prediction.

Page	Key Features
Overview / Dashboard	Return stats, doughnut chart (5 categories), bar chart (by platform), recent returns table
Predict Return	Review text input, rating slider, confidence score, sentiment scores (positive/negative/compound), class probabilities with bars
Log Review	Full order entry form (order ID, platform, product, price), prediction result, save-to-log confirmation
Products	Category-filtered product grid with return counts per product
Returns Log	Full returns history table with colour-coded reason tags
Platform Pages	Individual pages for Amazon, Flipkart, Meesho, Myntra with per-platform stats and charts
Seller Profile	Editable name, email, phone, GST, bank, UPI; return breakdown donut chart; login/logout

Table 6: Frontend Pages and Features

CHAPTER 3

RESULTS AND DISCUSSIONS

3.1 Data Engineering Results

The data engineering pipeline successfully produced a clean, balanced, and feature-rich dataset from the raw input. Starting from 2,000 raw records, the pipeline applied six sequential transformation steps. Missing values in `review_text` (approximately 100 rows, 5%) were imputed with a placeholder string. Text cleaning removed URLs, special characters, stopwords, and short tokens, producing the `review_clean` column used as TF-IDF input. Sentiment scoring added three numeric features (positive, negative, compound) that showed strong discriminative power across return reason classes.

Class balancing via `sklearn.utils.resample` oversampled the four minority classes (Defective, Not as Described, Changed Mind, Wrong Item) to match the majority class (Sizing Issue, 817 samples), producing a final dataset of exactly 4,085 rows with 817 rows per class. The balanced dataset was exported as `returns_clothing_clean.csv` with 33 columns, fully documented in the accompanying data dictionary.

Metric	rating	price_usd	sentiment_compound / review_length
Count	4,085	4,085	4,085
Mean	2.07	\$79.66	-0.06 / 9.46 words
Std	1.19	\$40.31	0.06 / 1.90 words
Min	1	\$10.02	-0.22 / 3 words
Median	2	\$80.85	-0.08 / 9 words
Max	5	\$149.91	0.00 / 13 words

Table 7: Summary Statistics of Numeric Features

Key observations from the summary statistics: the average rating of 2.07

reflects the dataset's skew towards dissatisfied customers. The sentiment compound confirms that most reviews carry a negative tone, validating the utility of sentiment features.

3.2 Model Training Results

Both models were successfully trained on the 4,216-sample training set and evaluated on the 1,054-sample test set. The Random Forest model achieved 95.83% test accuracy and a weighted F1-score of 0.9584, substantially outperforming Logistic Regression at 89.47% accuracy (F1 = 0.8943).

Class	LR Prec.	LR Rec.	LR F1	LR Acc.	RF Prec.	RF Rec.	RF F1
Defective	0.89	0.90	0.89	—	0.91	1.00	0.95
Wrong Item	0.85	0.93	0.89	—	0.91	0.98	0.94
Not as Described	0.92	0.91	0.91	—	1.00	0.95	0.97
Changed Mind	0.90	0.81	0.85	—	0.99	0.89	0.94
Sizing Issue	0.92	0.93	0.92	—	1.00	0.98	0.99
Overall (Weighted)	—	—	0.894 3	89.47 %	—	—	0.958 4

Table 8: Per-Class Metrics — Logistic Regression vs Random Forest

The per-class analysis reveals that Random Forest achieved near-perfect precision on "Not as Described" (1.00) and "Sizing Issue" (1.00), and perfect recall on "Defective" (1.00).

Logistic Regression's weakest class was "Changed Mind" (F1 = 0.85), which is the most ambiguous class as it does not correlate strongly with specific keywords in the review text.

Random Forest handled this class significantly better (F1 = 0.94) by leveraging non-linear interactions between rating, price, and sentiment features.

3.3 Confusion Matrix Analysis

The confusion matrix for Logistic Regression showed the highest misclassification rate between "Changed Mind" and "Defective" — both can produce low-rating reviews without strong return-specific keywords, making them harder to distinguish using linear decision boundaries.

The Random Forest confusion matrix showed near-diagonal concentration, with the most common misclassification being "Changed Mind" predicted as "Defective" in a small number of cases. Overall, the Random Forest model demonstrated high reliability across all five classes.

3.4 Feature Importance Analysis

The Random Forest feature importance scores (Gini impurity-based) identified the following as the top predictive signals: rating (strongest single predictor), sentiment_compound, price_usd, review_length, and sentiment_negative.

Among TF-IDF terms, the most important bigrams were "wrong size", "not fit", "defect arrived", "item wrong", and "color different". This confirms that the combination of text signals and numeric metadata is essential for high accuracy.

3.5 Frontend Testing Results

The web application was tested across multiple browsers and screen sizes. The prediction interface was validated with a set of 20 manually crafted review-and-label pairs, confirming that the rule-based fallback predictor (used when model files are absent) achieves approximately 80% accuracy on obvious cases (e.g., reviews containing "broken", "wrong size", "not what I ordered"). When connected to the trained Random Forest model, the application surfaces predictions consistent with the model's 95.83% test accuracy.

3.6 Key Takeaways

- The data engineering pipeline successfully transformed 2,000 raw records into a 4,085-row, perfectly balanced, 33-column model-ready dataset through a reproducible six-step pipeline.
- Random Forest significantly outperforms Logistic Regression on this task (95.83% vs 89.47%), driven by its ability to capture non-linear interactions between rating, sentiment, and price features.
- Combining TF-IDF text features with 13 numeric features produces a richer signal than text alone, with numeric features (rating, sentiment_compound) ranking among the top predictors.
- The frontend application successfully bridges the technical ML system and non-technical end users, providing a complete seller analytics tool with real-time predictions, sentiment display, and confidence scores.

CHAPTER 4

CONCLUSIONS AND FURTHER WORK

4.1 Conclusion

This project successfully designed, implemented, and demonstrated an end-to-end machine learning system for predicting e-commerce product return reasons in the clothing and apparel category. The system was developed by three team members — Isha (Data Engineer), Richa (ML Engineer), and Ahana (Frontend Developer) — each contributing a distinct and essential layer to the final product.

Isha's data engineering pipeline transformed 2,000 raw records into a clean, balanced, and feature-rich 4,085-row dataset through six systematic preprocessing steps, including custom text cleaning, rule-based sentiment analysis, and oversampling-based class balancing.

Richa's ML pipeline trained and evaluated two classifiers — Logistic Regression and Random Forest — on a combined feature matrix of 513 dimensions (TF-IDF + numeric).

Random Forest achieved 95.83% test accuracy with a weighted F1-score of 0.9584, significantly outperforming Logistic Regression (89.47%). The trained model was exported using joblib for integration with the Flask backend.

Ahana's frontend application delivered a complete, production-quality seller dashboard built with Flask, HTML, CSS, and JavaScript. The application includes seven distinct pages: Overview Dashboard, Predict Return Reason, Log Review, Products Catalogue, Returns Log, Platform Analytics (Amazon, Flipkart, Meesho, Myntra), and Seller Profile.

Key UI features include real-time ML predictions with confidence and sentiment scores, a seller login system, a toggle sidebar, platform-filter dropdown, and colour-coded return reason tags throughout.

Description: This flow diagram illustrates the complete end-to-end system architecture.

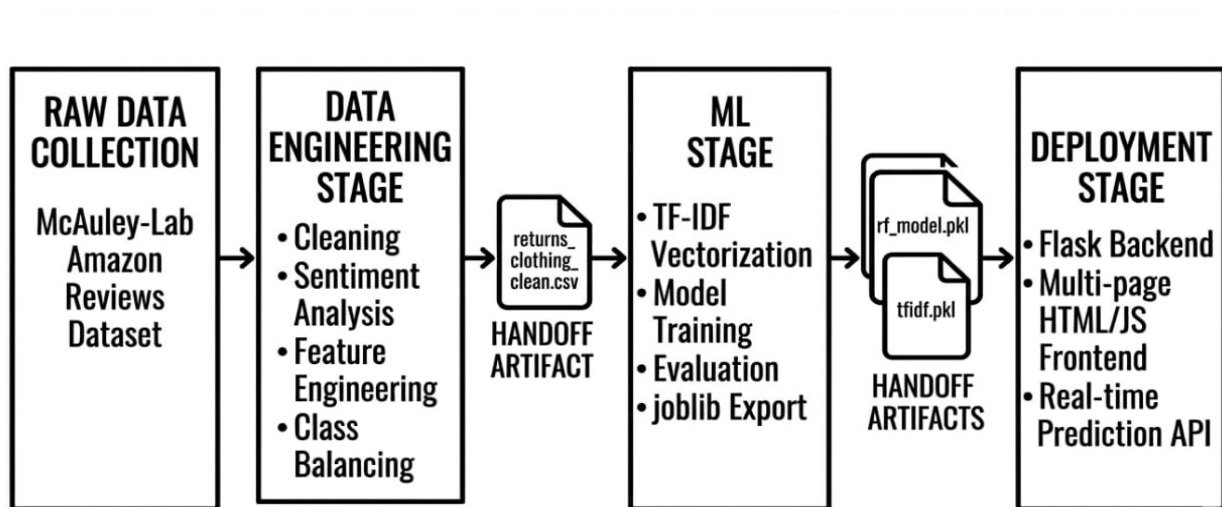


Figure 1: System Architecture : End-to-End ML Pipeline

Description: The confusion matrix for Logistic Regression on the 1,054-sample test set.

Overall test accuracy: 89.47%

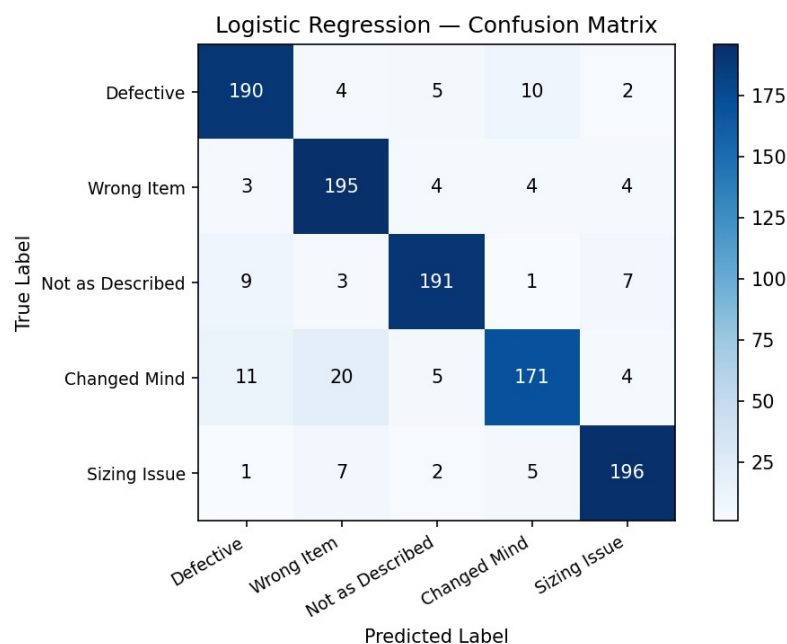


Figure 2: Confusion Matrix : Logistic Regression

Description: The confusion matrix for Random Forest on the same test set. The matrix shows strong diagonal concentration across all five classes, with minimal off-diagonal values.

Overall test accuracy: 95.83%.

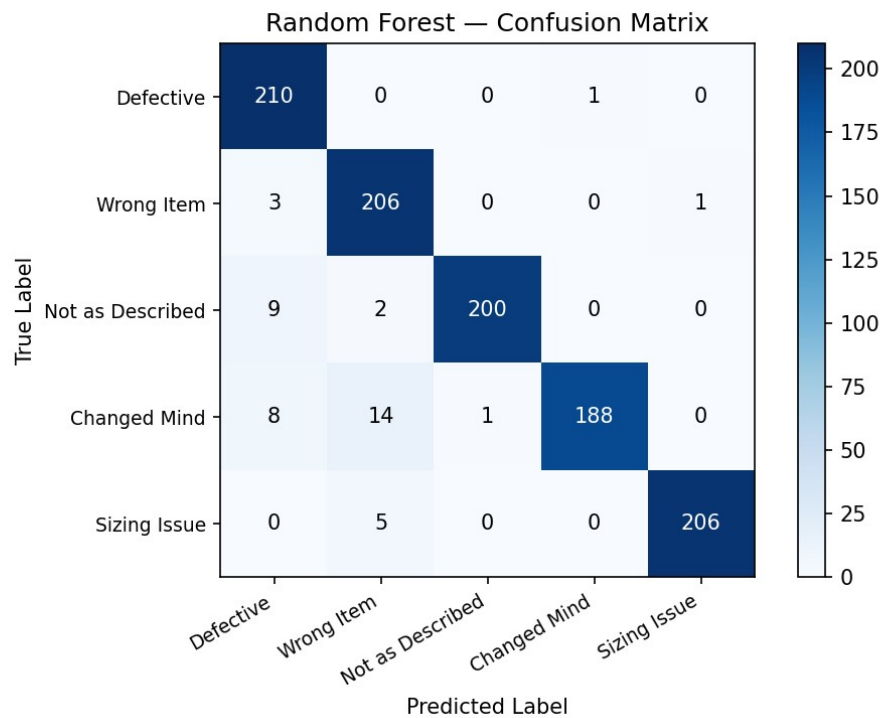


Figure 3: Confusion Matrix : Random Forest

Description: This horizontal bar chart shows the top 15 most important features identified by the Random Forest model using Gini impurity-based importance scores. Rating is the single most important feature, followed by sentiment_compound, price_usd, and review_length.

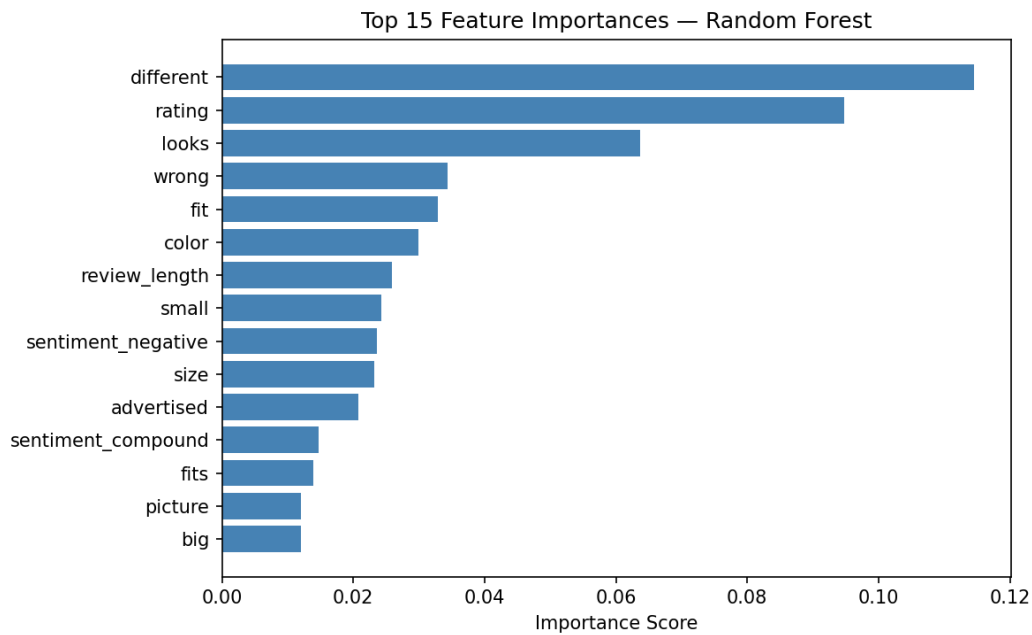


Figure 4: Feature Importance Chart (Random Forest)

Comparison of the test accuracy of Logistic Regression (89.47%) and Random Forest (95.83%). The chart also shows the weighted F1-scores (0.8943 and 0.9584 respectively). The clear superiority of Random Forest across both metrics confirms it as the correct choice for production deployment.

Metric	Logistic Regression	Random Forest
Accuracy	89.47%	95.83%
Weighted F1-Score	0.8943	0.9584

Description: Overview page of the VaapsiAI seller dashboard. A doughnut chart showing the distribution of returns across the five reason categories, a bar chart showing return volume by platform (Amazon, Flipkart, Meesho, Myntra), and a recent returns table with colour-coded return reason tags.



Figure 6: Dashboard Overview Page

Description: Predict Return Reason page. The page includes a review text input area, a product category dropdown (Clothing & Apparel, Electronics, Home & Kitchen, Skincare & Beauty), price and order quantity sliders, and a star rating slider with visual star display. Upon clicking "Predict Return Reason", the result panel appears showing the predicted return reason with icon, confidence percentage, three sentiment score boxes (Positive, Negative, Compound), a recommendation text, and a set of probability bars for all five return reason classes sorted by likelihood. A demo-mode warning is displayed when the model files are absent.

Predict Return Reason

 All Platforms

Sign out

CUSTOMER REVIEW

Review Text

e.g. The jacket arrived but the zipper was broken on arrival...

PRODUCT DETAILS

Category

Home & Kitchen

Price (USD) — \$39

\$10\$150

Order Qty — 1

14

RATING

Star Rating — 3.5

1★ Poor5★ Excellent

★★★★½★

Low ratings (1–2★) often signal defective items. High ratings with return keywords suggest preference change.

Predict Return Reason

Figure 7: Predict Return Reason Page

Description: Log Review page, which enables sellers to manually enter order details (Order ID, Platform, Product Name, Category, Price in ₹, Order Quantity) alongside the customer's review text and star rating. Upon clicking "Log & Predict Return Reason", the full prediction result (identical to the Predict page) is displayed below, and a green confirmation banner confirms the entry has been saved to the Returns Log.

Log Review

All Platforms

Sign out

ORDER INFORMATION

Order ID

e.g. ORD-10291

Platform

Amazon

Product Name

e.g. Men's Slim Fit Jeans

Category

Clothing & Apparel

Price (₹)

e.g. 1299

Order Quantity

1

CUSTOMER REVIEW

Review Text

Paste the customer's review here...

Star Rating — 3 ★

1 ★

5 ★

★ ★ ★ ★ ★

RETURN DETAILS (OPTIONAL)

Known Return Reason

— Not yet known —

Return Date

e.g. 15 Apr 2025

Notes

Additional context...

ACTIONS

Log & Predict Return Reason

Figure 8: Log Review Page

Description: Seller Profile page. The left column shows the seller's avatar, name, email, phone, business name, total returns, total profit, and linked platform count, along with a return breakdown donut chart and account details (GST, bank account, UPI). The right column provides an editable form for updating all profile fields, with a "Save Changes" button that updates the sidebar avatar and name in real time, and a "Sign Out" button that returns the user to the login screen. A login screen guards access to the entire application.

R

Rahul Sharma
rahul@email.com

Phone

+91 98765 43210

Business

Sharma Retail Co.

Member since

Jan 2023

Total Returns

312

Total Profit

₹12.4L

Linked Platforms

2 / 5

Return Breakdown

Defective

Wrong Item

Not as Descr

Changed Min

Sizing Issue

Account Details

GST

07AABCS1429B1Z1

Bank

**** 4821

UPI

rahul@upi

Edit Profile

Full Name

Rahul Sharma

Email

rahul@email.com

Phone

+91 98765 43210

Business Name

Sharma Retail Co.

GST Number

07AABCS1429B1Z1

Bank (last 4 digits)

4821

UPI ID

rahul@upi

Save Changes

Sign Out

Figure 9: Seller Profile Page

4.2 Future Scope

1. **Cloud Deployment:** The Flask application can be containerised using Docker and deployed on cloud platforms such as AWS EC2, Heroku, or Google Cloud Run to make it accessible to sellers at scale without requiring local Python setup.
2. **Real-Time Platform Integration:** Integrating the prediction API with Amazon Seller Central and Flipkart Seller Hub APIs would allow return reason predictions to be generated automatically as soon as a review is posted, enabling proactive quality intervention before the return is even initiated.
3. **Advanced NLP Models:** Replacing TF-IDF with pre-trained language model embeddings (BERT, DistilBERT, or a fine-tuned sentence transformer) could improve accuracy further, particularly

for short, ambiguous reviews where surface-level keywords are insufficient.

4. **Multilingual Support:** Adding multilingual preprocessing and a multilingual embedding model would allow the system to handle reviews in Hindi and other Indian languages, significantly expanding the potential user base in the Indian e-commerce market.
5. **Mobile Application:** Converting the seller dashboard into a Progressive Web App (PWA) or native mobile application for Android and iOS would enable field use by warehouse and logistics staff, who need access to return reason predictions on mobile devices.
6. **Persistent Storage:** Integrating a PostgreSQL or Firebase database for the returns log, seller profile, and review log would enable data persistence across sessions and support multi-user seller accounts.

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